DISASTER MITIGATION & RESILIENCE

BeaconPath

Zero-Internet Emergency Mesh & Dynamic Hazard-Aware Evacuation Navigation

Core Paradigm
100% Off-Grid Peer-to-Peer

Target Impact
Eliminate The Blackout Trap

Stack
BLE, Wi-Fi Direct & Offline Vector GIS

THE PROBLEM

The Blackout Trap

70%–90%

Cellular tower collapse within 6 hours.

Total Blackout: Civilian calls fail instantly.

Blind Evacuation: Online maps show no connection.

Submerged Hazards: Fleeing families enter danger zones.



CORE PARADIGM

Introducing BeaconPath

PILLAR 01

Decentralized P2P Mesh

Multi-hop smartphone relay network.

PILLAR 02

On-Device Safe Routing

Dynamic offline hazard bypass.

OPERATIONAL WORKFLOW

Field Relay Pipeline

01

Satellite GPS Fix

Pre-cached 80MB city vector map.

02

Hazard Broadcast

One-tap micro-packet incident beacon.

03

Multi-Hop Reroute

Silent peer propagation for detours.

LOCAL COMPUTATION

Real-Time Recalibration

< 50ms

On-Device Graph Recalculation Speed

Instant offline detour calculation using A* graph algorithms.

ENGINEERING

Lightweight Technical Stack

Presentation

Flutter and React Native client.

Offline Cache

SQLite and MBTiles format.

Mesh Layer

Google Nearby and BLE protocol.

Path Engine

GraphHopper on-device search.

EFFICIENCY PROFILE

Designed for Emergency Survival

< 3%

Battery Drain / Hour

Continuous background BLE

< 80MB

City Map Footprint

Compressed vector tiles

0 B

Internet Bandwidth

Full offline functionality

BENCHMARK

Competitive Advantage

Capability	Google / Apple Maps	Standard SOS Apps	BeaconPath
Zero-Internet / No SIM	No	No	**100% Offline**
P2P Mesh Hop Relay	No	No	**Yes (BLE & Wi-Fi)**
Hazard Detour Routing	Cloud-Dependent	None	**Instant On-Device**
Triage Heatmap Relay	No	Partial (Server)	**Autonomous Mesh**

FUTURE EXPANSION

Extending the Physical Mesh



$15 LoRaWAN Infrastructure
Long-distance relay up to 10km.

Autonomous Aerial Drones
Airborne harvesting of SOS packets.

Municipal Agency Link
Direct integration with rescue command.

NETWORK TRUST

Decentralized Verification

The Spoofing Attack

Malicious actors injecting fake flooded road reports to manipulate civilian movement.

Consensus Guard

Requires multi-device signatures or cryptographically verified emergency responders.



”BeaconPath turns every citizen into a beacon of resilience, ensuring that when the grid goes down, humanity stays connected.”

BeaconPath Disaster Resilience System

BeaconPath

Open for questions from the judging panel

100% Offline • BLE / Wi-Fi Mesh • Hazard-Aware Rerouting

Image Sources



https://www.ipsnews.net/Library/2025/12/Gampaha_.jpg

Source: www.ipsnews.net



https://cdn.sanity.io/images/vjj9kuff/production/e87e12ff47ccbd85d78d9af228f244f0acb1a60d-1500×918.jpg?q=85&auto=format&w=1920

Source: flytbase.com